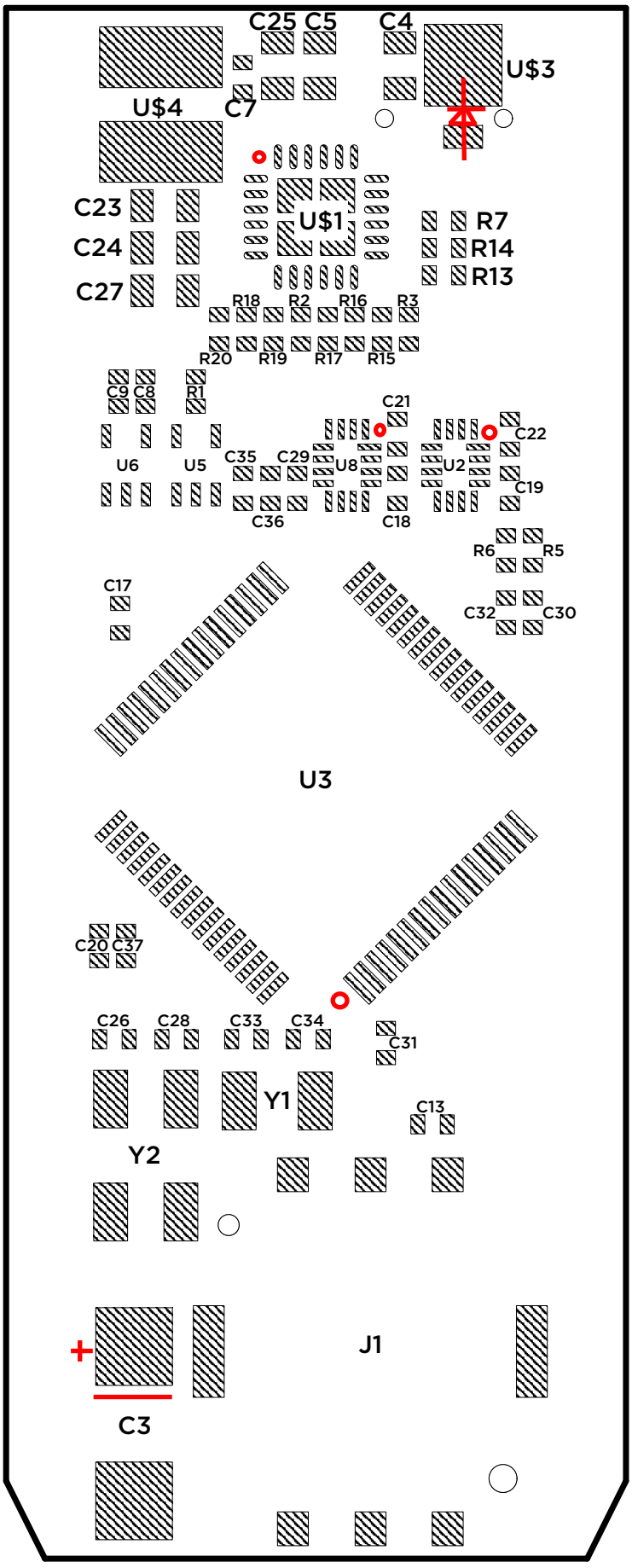ELECTRON - BOTTOM